WORKED SOLUTION

STEP II 2022, Question 7

Pure • Roots of Complex Polynomials

STEP II 2022, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

The equation $\mathrm{h}(w) = 0$ splits naturally as $w^5(nw+1) = -(w+n)$, and taking moduli turns it into a statement about $|w|$ alone. Once every root is known to lie on the unit circle, each non-real root pairs with its conjugate, which is also its reciprocal. That gives the quadratic factors, and after that it's just comparing coefficients and a sign argument.

Solution

Part (i)

(a) Group the terms of $\mathrm{h}(w) = 0$ as

$$w^5(nw+1) = -(w+n).$$

Neither bracket can vanish. If $w = -1/n$ then $\mathrm{h}(w) = n - \frac{1}{n} \neq 0$, and if $w = -n$ then $\mathrm{h}(w) = n^5(n^2-1) \neq 0$, since $n \geq 2$. Taking moduli and squaring,

$$|w^5|^2\,|nw+1|^2 = |w+n|^2.$$

For any complex u and real c we have $|u+c|^2 = (u+c)(\bar{u}+c) = |u|^2 + 2c\,\mathrm{Re}(u) + c^2$. So

$$|w+n|^2 = |w|^2 + 2n\,\mathrm{Re}(w) + n^2 = \mathrm{f}(w),$$
$$|nw+1|^2 = n^2|w|^2 + 2n\,\mathrm{Re}(w) + 1 = \mathrm{g}(w).$$

Both are positive, so $|w^5|^2 = \mathrm{f}(w)/\mathrm{g}(w)$. Taking the positive square root gives $|w^5| = \sqrt{\mathrm{f}(w)/\mathrm{g}(w)}$, as required.

(b) The $\mathrm{Re}(w)$ terms cancel in the difference:

$$\mathrm{f}(w) - \mathrm{g}(w) = n^2 + |w|^2 - n^2|w|^2 - 1 = (n^2-1)(1-|w|^2).$$

Suppose, for contradiction, that $|w| < 1$. Since $n^2 - 1 > 0$, we get $\mathrm{f}(w) > \mathrm{g}(w) > 0$, so $\mathrm{f}(w)/\mathrm{g}(w) > 1$. By (a), $|w|^5 > 1$, so $|w| > 1$, which contradicts $|w| < 1$. Hence $|w| \geq 1$.

(c) The same argument works the other way round. If $|w| > 1$, then $\mathrm{f}(w) < \mathrm{g}(w)$, so $|w|^5 < 1$ and $|w| < 1$, a contradiction. So $|w| \leq 1$ as well, and together with (b), $|w| = 1$.

Ask yourself Is there a slicker reason why $|w| \leq 1$?

Yes. The coefficients $n, 1, 0, 0, 0, 1, n$ read the same backwards, so $z^6\,\mathrm{h}(1/z) = \mathrm{h}(z)$. Hence if w is a root, so is $1/w$, and applying (b) to $1/w$ gives $|w| \leq 1$.

Part (ii)

(a) Since h has real coefficients, its non-real roots come in conjugate pairs. The six roots are distinct and non-real, so they form three pairs $w_k,\ \bar{w}_k$ $(k = 1, 2, 3)$. By part (i), $|w_k| = 1$, so $w_k\bar{w}_k = 1$, and

$$(z - w_k)(z - \bar{w}_k) = z^2 - (w_k + \bar{w}_k)z + w_k\bar{w}_k = z^2 - a_k z + 1,$$

where $a_k = 2\,\mathrm{Re}(w_k)$ is real. The polynomial h has degree 6 and leading coefficient n, so

$$\mathrm{h}(z) = n(z^2 - a_1 z + 1)(z^2 - a_2 z + 1)(z^2 - a_3 z + 1).$$

(b) The z^5 terms of the product arise by taking $-a_k z$ from one factor and z^2 from the other two, so the coefficient of z^5 is $-n(a_1 + a_2 + a_3)$. In h that coefficient is 1, so

$$a_1 + a_2 + a_3 = -\frac{1}{n}.$$

(c) First multiply two of the factors:

$$(z^2 - a_1 z + 1)(z^2 - a_2 z + 1) = z^4 - (a_1 + a_2)z^3 + (2 + a_1 a_2)z^2 - (a_1 + a_2)z + 1.$$

Multiplying by $z^2 - a_3 z + 1$, the z^3 terms come from $z^3 \times 1$, $z^2 \times (-a_3 z)$ and $z \times z^2$:

$$-(a_1 + a_2) - a_3(2 + a_1 a_2) - (a_1 + a_2) = -2(a_1 + a_2 + a_3) - a_1 a_2 a_3.$$

There is no z^3 term in h, so $a_1 a_2 a_3 = -2(a_1 + a_2 + a_3)$, and by (b),

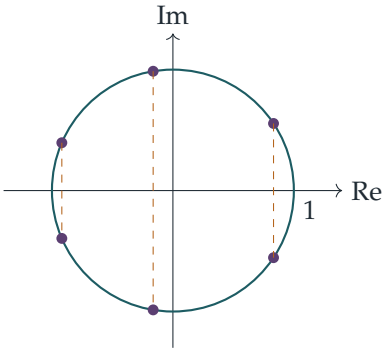
$$a_1 a_2 a_3 = \frac{2}{n}.$$

(d)

Ask yourself How do the signs of the a_k tell us about real parts?

Each conjugate pair $w_k, \bar{w}_k$ shares the real part $\mathrm{Re}(w_k) = a_k/2$. So the number of roots with negative real part is twice the number of negative a_k.

Since $a_1 a_2 a_3 = 2/n > 0$, none of the a_k is zero, and an even number of them are negative: none or two. They can't all be positive, because $a_1 + a_2 + a_3 = -1/n < 0$. So exactly two of the a_k are negative and one is positive, and **four** of the six roots have negative real part.



The figure shows the case $n = 2$. The six roots lie on the unit circle in three conjugate pairs, and two of the pairs are to the left of the imaginary axis.

Checks

- As a sanity check that no root is real, note that $\mathrm{h}(1) = 2n + 2$ and $\mathrm{h}(-1) = 2n - 2$ are both non-zero. Since the roots lie on the unit circle, ± 1 are the only real candidates.
- For $n = 2$, solving numerically gives $a_k \approx -1.8368,\ -0.3272,\ 1.6640$. These sum to $-0.5 = -\frac{1}{2}$, their product is $1.000 = \frac{2}{2}$, and two are negative, matching (b), (c) and (d). A computer check of $n = 2, \ldots, 29$ confirmed that every root has modulus 1 and that four roots have negative real part each time.

Common slips

- Dividing by $nw + 1$ without checking that $w = -1/n$ isn't a root.
- In (c), missing one of the three ways to make z^3, especially the term $-2a_3$ that comes from the constant 2 in $2 + a_1 a_2$.
- In (d), counting negative a_k but forgetting that each one accounts for *two* roots.

Beyond STEP

A polynomial whose coefficients read the same backwards is called *palindromic* (or self-reciprocal). Its roots come in pairs $w, 1/w$, and the substitution $u = z + 1/z$ halves its degree. Here $a_k = w_k + 1/w_k$ is exactly that u. Questions about when all the roots lie on the unit circle lead to results such as the Schur–Cohn test and to Rouché's theorem in complex analysis. Needham's *Visual Complex Analysis* is a beautiful place to see these ideas geometrically.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.